

Rakib Hasan

rakibhasan1107@gmail.com | (+880) 1628-546970 | Narsingdi, Bangladesh | [linkedin.com/in/rakib-hasan-797b07228](https://www.linkedin.com/in/rakib-hasan-797b07228)

Ph.D. Applicant, Fall 2027 – Technical HCI · AR/VR · AI/ML for Interactive Systems

SUMMARY

Software engineer with 3+ years at Samsung R&D Institute Bangladesh building real-time smart home interfaces on Windows, with hands-on experience in Unity and a VR proof-of-concept for controlling home appliances. Strong programmer in Python, PyTorch, C#, and C++ and a competitive programmer, with an AI/ML background, two peer-reviewed publications on modeling under uncertainty, and patent contributions. Seeking to develop and evaluate AI models of user behavior and adaptive, immersive systems.

RESEARCH INTERESTS

Human-Computer Interaction · AR/VR & Immersive Interaction · Smart Home & IoT Interfaces · AI/ML Models of User Behavior · Adaptive and Accessible UI · Uncertainty Modeling

EDUCATION

B.Sc. in Computer Science & Engineering – Khulna University of Engineering & Technology

2018 – 2023

CGPA: 3.12 / 4.00 | Khulna, Bangladesh | AI/ML coursework and PyTorch-based projects

- **Thesis:** Risk Evaluation of Explosive, Flammable, and Toxic Chemicals Using a Fuzzy Inference System – built a fuzzy-logic model to assess hazard vulnerability and precaution levels under uncertain inputs.

INDUSTRY EXPERIENCE

Senior Software Engineer (Research & Platform) – Samsung R&D Institute Bangladesh

Mar 2025 – Aug 2026

- Led architecture and end-to-end implementation of **SmartThings Routine Creation & Editing** on Windows: a real-time UI (SSE, async/await, MVVM) for automating lights, thermostats, music, and other devices; shipped on time and recognized by leadership.
- Supported the SmartThings Windows feature integrating **Unity** for an interactive **3D map view of the home**.
- Worked hands-on on a **VR proof-of-concept** for controlling SmartThings-connected appliances (e.g., turning lights on/off) from VR, demonstrated to headquarters visitors at Samsung R&D Bangladesh.
- Proposed and prototyped AI-based PoCs for the smart home platform (pitch accepted by leadership); contributed patent ideas and Mobile Developer Conference (MDC) submissions.
- Aligned Windows behavior with the Android counterpart across global teams; mentored junior engineers on code quality and architecture.

Software Engineer (Platform Development) – Samsung R&D Institute Bangladesh

Apr 2023 – Feb 2025

- Built live camera streaming in SmartThings AVPlayer with **GStreamer and WebRTC**, integrated into WPF; evaluated multiple architectures before the final design.
- Contributed to Samsung DeX Windows features (file transfer, localization, **C++ integration**).
- Compared .NET UWP, WPF, and WinUI 3 through demo apps for the Nearby Device project to guide the technology-stack decision; contributed smart home PoCs and patent ideation.

SELECTED PROJECTS

- **Dots & Boxes with AI Opponent (Unity/C#, Undergraduate Project)** – complete game with player-vs-player and player-vs-AI modes, configurable grid size and difficulty levels, and a full Unity UI (menus, settings, tutorial, live scoreboard, replay); AI built on game-tree search with state memoization. github.com/blueshark-14/dots-boxes-unity-ai
- **Crop Prediction & Disease Detection System** – ML system that recommends crops from climate/soil data and classifies crop diseases.
- **Risk Evaluation via Fuzzy Inference System (Thesis)** – modeled chemical hazard vulnerability under uncertainty; led to the publications below.

PROGRAMMING CONTESTS & ALGORITHMIC FOUNDATIONS

- **1st Runner-Up, KU CSE Fest Programming Contest (2022)**. Proficient in algorithms, data structures, dynamic programming, number theory, and probability.
- Passed Samsung Software Competency Test (Professional Level); earned the Excellence in SW Certificate (2023).

PUBLICATIONS & PATENT

Journal: A. Aziz, M. M. Suzon, **R. Hasan**, "A fuzzy logic-based risk evaluation and precaution level estimation of explosive, flammable, and toxic chemicals for preventing damages," *Heliyon*, vol. 11, no. 1, e41216, 2025.
doi.org/10.1016/j.heliyon.2024.e41216

Conference: M. M. Suzon, **R. Hasan**, A. Aziz, A. Z. M. N. Abir, "Risk Evaluation of Explosive and Flammable Chemicals Using Fuzzy Inference System," in *Proc. 2nd International Conference on Big Data, IoT and Machine Learning (BIM 2023)*, Lecture Notes in Networks and Systems, vol. 867, Springer, Singapore, 2024, pp. 911–921.
link.springer.com/chapter/10.1007/978-981-99-8937-9_60

Patent Contributions: Contributed several smart home automation invention ideas at Samsung R&D; some passed Samsung's initial internal patent review.

TECHNICAL SKILLS

Languages & ML: Python, PyTorch, scikit-learn, pandas, C++, C, C#, Kotlin, Java, JavaScript

AR/VR & Interactive Systems: Unity (game UI and AI logic; 3D home-map integration support), VR PoC development

Platforms & Frameworks: .NET (UWP, WPF, WinUI 3), MVVM, SmartThings SDK, Samsung DeX SDK

Systems & Protocols: GStreamer, WebRTC, SSE, async/event-driven design, Git, CI/CD, MySQL

Research Tools: LaTeX, fuzzy logic systems, academic writing

HONORS & AWARDS

- **Icon Award**, Jan–Feb 2026 – Samsung R&D (2026)
- Passed Samsung Basic **AI Certification Exam** – Samsung R&D (2025)
- Excellence Award for Outstanding Contribution – Samsung R&D (2024)
- Excellence in **SW Certificate Test** – Samsung R&D (2023)
- 1st Runner-Up – KU CSE Fest Programming Contest (2022)